Optimal Runtime of Noisy Amplitude Amplification from Spectral Stability

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Abstract

We study amplitude amplification under per-iteration noise modeled by a completely positive trace-preserving contraction acting after each Grover iterate. Working in the Liouville representation, we analyze the spectrum of the induced noisy superoperator and show that the ideal Grover eigenstructure is stable up to perturbations of order $O(p)$, where p is the noise strength per step. This yields a damped-rotation picture for the success probability and leads to an optimal stopping time

$$T^* \asymp \min\left(\frac{1}{\theta}, \frac{1}{p}\right),$$

where $\theta \asymp 1/\sqrt{N}$ is the Grover rotation angle. As a consequence, the quadratic speedup survives only when $p \ll 1/\sqrt{N}$, while the regime $p = \Omega(1/\sqrt{N})$ marks the onset of noise-limited, effectively classical behavior.

1 Introduction

Grover's search algorithm achieves a quadratic quantum speedup for unstructured search, reducing query complexity from $O(N)$ to $O(\sqrt{N})$ [1]. The algorithm proceeds by repeated application of the Grover iterate, which acts as a rotation on the unit circle within a two-dimensional invariant subspace spanned by the marked state and its orthogonal complement. In the ideal noiseless case, the dynamics is governed by a unit-modulus eigenvalue pair $e^{\pm i\theta}$, producing a coherent rotation along the unit circle. After $O(\sqrt{N})$ iterations, the state is rotated close to the marked state and measurement succeeds with high probability.

In practice, each Grover step is affected by noise. The evolution is therefore described not by a unitary iterate but by a quantum channel $\mathcal{G} = \mathcal{N} \circ \mathcal{U}_G$, where $\mathcal{U}_G$ denotes the ideal Grover channel and $\mathcal{N}$ is a completely positive trace-preserving (CPTP) map. Because Grover's algorithm requires many iterations, even weak per-step noise can accumulate and alter the amplification mechanism. This raises a natural question: how stable is amplitude amplification under repeated perturbations, and at what noise level does the quadratic speedup break down?

2 Noisy Model

2.1 Noisy Grover Iterate as a CPTP Perturbation

Let

$$G = DO \tag{2.1}$$

denote the ideal Grover iterate, where O is the oracle and D is the diffusion operator. In the density-matrix representation, the ideal iterate is described by the unitary channel

$$\mathcal{U}_G(\rho) = G\rho G^\dagger. \tag{2.2}$$

To model noise, we consider a CPTP channel $\mathcal{N}_p$ and define the noisy iterate by

$$\mathcal{G}_p = \mathcal{N}_p \circ \mathcal{U}_G. \tag{2.3}$$

We assume that the noise is a small perturbation of the identity channel:

$$\mathcal{N}_p = (1-p)\,\mathrm{id} + p\,\mathcal{E}, \qquad p \in [0,1], \tag{2.4}$$

where $\mathcal{E}$ is a fixed CPTP map. Equivalently,

$$\mathcal{G}_p = (1-p)\,\mathcal{U}_G + p\,\mathcal{E} \circ \mathcal{U}_G. \tag{2.5}$$

Thus each Grover step is replaced by a convex mixture of the ideal evolution and a perturbed evolution. After T iterations, the state evolves as

$$\rho_T = \mathcal{G}_p^T(\rho_0). \tag{2.6}$$

Since the ideal algorithm requires $T = O(\sqrt{N})$ steps, the cumulative effect of per-step noise becomes significant precisely when pT is no longer small. This heuristic already suggests that the natural threshold is

$$p \asymp \frac{1}{\sqrt{N}}, \tag{2.7}$$

2.2 Liouville Representation

To analyze repeated noisy evolution, it is convenient to pass to the Liouville representation. Under vectorization,

$$\rho \mapsto |\rho\rangle\rangle, \tag{2.8}$$

a quantum channel becomes a linear map on a d^2-dimensional operator space:

$$|\rho'\rangle\rangle = \mathbf{G}_p|\rho\rangle\rangle. \tag{2.9}$$

For a unitary channel $\mathcal{U}_G(\rho) = G\rho G^\dagger$, the corresponding Liouville matrix is

$$\mathbf{U}_G = G \otimes \overline{G}, \tag{2.10}$$

where $\overline{G}$ denotes complex conjugation in the computational basis. The noisy iterate takes the form

$$\mathbf{G}_p = \mathbf{N}_p\mathbf{U}_G. \tag{2.11}$$

3 Spectral Stability of the Noisy Iterate

We now establish two structural facts. First, the noisy iterate is a contraction, so its spectrum is confined to the closed unit disk. Second, when the noise is a small CPTP perturbation of the identity, the spectrum of the noisy iterate remains within $O(p)$ of the ideal spectrum.

3.1 Contraction and Spectral Radius Bound

Let

$$\|\mathcal{T}\|_{1\to 1} := \sup_{X\neq 0} \frac{\|\mathcal{T}(X)\|_1}{\|X\|_1} \tag{3.1}$$

denote the induced trace-norm superoperator norm.

Theorem 1 (CPTP contraction and spectral radius bound)**.** *Let Φ be a CPTP map. Then:*

1. *Φ is contractive in trace norm:*

$$\|\Phi(X)\|_1 \leq \|X\|_1 \qquad \textit{for all operators } X.$$

Equivalently,

$$\|\Phi\|_{1\to 1} \leq 1.$$

2. Every eigenvalue λ of Φ satisfies

$$|\lambda| \leq 1.$$

In particular, the noisy Grover iterate $\mathcal{G}_p$ is a contraction, and its Liouville spectrum lies in the closed unit disk [2, 3].

Proof. We first prove trace-norm contractivity. It is enough to consider Hermitian X, since the general case follows by decomposing X into real and imaginary parts.

Let $X = X^\dagger$, and write its Jordan decomposition as

$$X = X_+ - X_-, \qquad X_\pm \succeq 0, \qquad X_+ X_- = 0.$$

Then

$$\|X\|_1 = \mathrm{Tr}(X_+) + \mathrm{Tr}(X_-).$$

Since Φ is positive, $\Phi(X_\pm) \succeq 0$, and since Φ is trace-preserving,

$$\mathrm{Tr}(\Phi(X_\pm)) = \mathrm{Tr}(X_\pm).$$

Therefore,

$$\begin{aligned}
\|\Phi(X)\|_1 &= \|\Phi(X_+) - \Phi(X_-)\|_1 \\
&\leq \|\Phi(X_+)\|_1 + \|\Phi(X_-)\|_1 \\
&= \mathrm{Tr}(\Phi(X_+)) + \mathrm{Tr}(\Phi(X_-)) \\
&= \mathrm{Tr}(X_+) + \mathrm{Tr}(X_-) \\
&= \|X\|_1.
\end{aligned}$$

Hence $\|\Phi\|_{1\to 1} \leq 1$.

Now let $X \neq 0$ be an eigen-operator with

$$\Phi(X) = \lambda X.$$

Then

$$|\lambda|\, \|X\|_1 = \|\lambda X\|_1 = \|\Phi(X)\|_1 \leq \|X\|_1,$$

so $|\lambda| \leq 1$.

Since $\mathcal{G}_p$ is a composition of CPTP maps, it is itself CPTP, and the conclusions apply directly. □

Theorem 1 gives the basic qualitative picture: noise cannot enlarge the Grover spectrum, only contract it. Thus the ideal unit-circle rotation becomes a damped rotation once noise is introduced.

3.2 Spectral Stability Under Small CPTP Perturbations

We next quantify how far the noisy spectrum can move from the ideal one when the perturbation is weak.

Theorem 2 (Spectral stability under CPTP perturbation)**.** *Let*

$$\mathcal{G}_p = \mathcal{N}_p \circ \mathcal{U}_G, \qquad \mathcal{N}_p = (1-p)\,\mathrm{id} + p\,\mathcal{E},$$

where $\mathcal{E}$ is CPTP, and define

$$\Delta_p := \mathcal{G}_p - \mathcal{U}_G.$$

Then

$$\|\Delta_p\|_{1\to 1} \le 2p.$$

Consequently, every eigenvalue λ of $\mathcal{G}_p$ satisfies

$$\mathrm{dist}\big(\lambda, \sigma(\mathcal{U}_G)\big) \le \|\Delta_p\|_{1\to 1} \le 2p.$$

Thus the spectrum of the noisy iterate lies within a $2p$-neighborhood of the ideal Grover spectrum in the complex plane.

Proof. We compute

$$\Delta_p = \mathcal{N}_p \circ \mathcal{U}_G - \mathcal{U}_G = (\mathcal{N}_p - \mathrm{id}) \circ \mathcal{U}_G = p(\mathcal{E} - \mathrm{id}) \circ \mathcal{U}_G.$$

Taking the induced trace norm and using submultiplicativity,

$$\|\Delta_p\|_{1\to 1} \le p\,\|\mathcal{E} - \mathrm{id}\|_{1\to 1}\,\|\mathcal{U}_G\|_{1\to 1}.$$

Since unitary conjugation preserves trace norm,

$$\|\mathcal{U}_G\|_{1\to 1} = 1.$$

Moreover, by Theorem 1,

$$\|\mathcal{E}\|_{1\to 1} \le 1 \qquad \text{and} \qquad \|\mathrm{id}\|_{1\to 1} = 1.$$

Hence

$$\|\mathcal{E} - \mathrm{id}\|_{1\to 1} \le \|\mathcal{E}\|_{1\to 1} + \|\mathrm{id}\|_{1\to 1} \le 2,$$

which gives

$$\|\Delta_p\|_{1\to 1} \le 2p.$$

For the spectral inclusion, let $z \notin \sigma(\mathcal{U}_G)$. Let $\sigma_\varepsilon(A)$ denote the ε-pseudospectrum of A, defined by [4]

$$\sigma_\varepsilon(A) = \left\{ z : \|(zI - A)^{-1}\|_{1\to 1} \geq \frac{1}{\varepsilon} \right\}.$$

If

$$z \notin \sigma_{\|\Delta_p\|_{1\to 1}}(\mathcal{U}_G),$$

then

$$\|(zI - \mathcal{U}_G)^{-1}\|_{1\to 1} \, \|\Delta_p\|_{1\to 1} < 1.$$

Hence

$$zI - \mathcal{G}_p = (zI - \mathcal{U}_G)\Big(I - (zI - \mathcal{U}_G)^{-1}\Delta_p\Big)$$

is invertible by the Neumann series argument, so $z \notin \sigma(\mathcal{G}_p)$. Therefore

$$\sigma(\mathcal{G}_p) \subset \sigma_{\|\Delta_p\|_{1\to 1}}(\mathcal{U}_G).$$

Since $\mathcal{U}_G$ is unitary, it is normal, and for normal operators the ε-pseudospectrum coincides with the ε-neighborhood of the spectrum. Hence, for every $\lambda \in \sigma(\mathcal{G}_p)$,

$$\operatorname{dist}\big(\lambda, \sigma(\mathcal{U}_G)\big) \leq \|\Delta_p\|_{1\to 1}.$$

□

4 Optimal Runtime Under Noise

In the noiseless case, the success probability is

$$P_0(T) = \sin^2\big((T + \tfrac{1}{2})\theta\big), \tag{4.1}$$

which arises from a pair of unit-modulus eigenvalues $e^{\pm i\theta}$ governing the two-dimensional invariant subspace. By Theorem 2, the noisy iterate is an $O(p)$ perturbation of the ideal Grover superoperator, so the conjugate pair admits a perturbative expansion. Under weak noise,

$$\lambda_\pm(p) = (1 - cp + O(p^2))e^{\pm i(\theta + O(p))}, \qquad c > 0, \tag{4.2}$$

After T iterations, the coherent contribution is attenuated by

$$|\lambda_\pm(p)|^T \approx (1 - cp)^T = e^{-cpT + O(p^2 T)}. \tag{4.3}$$

Noise therefore exponentially damps the Grover oscillation while preserving its functional form to leading order. Assuming a spectral gap, the dynamics is dominated by the leading conjugate pair, yielding the damped-rotation approximation

$$P(T) \approx (1-cp)^T P_0(T) \approx e^{-cpT} \sin^2\big((T+\tfrac{1}{2})\theta\big)\,. \tag{4.4}$$

Theorem 3 (Optimal stopping time under noise)**.** *Let $\theta \asymp 1/\sqrt{N}$ be the Grover rotation angle and let p be the per-step noise strength. Under the damped-rotation approximation, the success probability is maximized at a stopping time*

$$T^* \asymp \min\left(\frac{1}{\theta}, \frac{1}{p}\right).$$

In particular:

1. *If $p \ll \theta$, then*
$$T^* \asymp \frac{1}{\theta} \sim \sqrt{N},$$
and quadratic scaling is preserved.

2. *If $p \gg \theta$, then*
$$T^* \asymp \frac{1}{p},$$
so the algorithm is stopped by noise before full Grover amplification can occur.

Proof sketch. The oscillatory factor peaks on the scale $T \sim 1/\theta$, while the damping e^{-cpT} decays on the scale $T \sim 1/p$. The maximum therefore occurs at the smaller of these two timescales.

If $p \ll \theta$, damping is negligible over the Grover runtime and quadratic scaling is preserved. If $p \gg \theta$, decay dominates before the first peak, limiting the useful depth to $T \sim 1/p$. □

5 Conclusion

We presented a spectral framework for noisy amplitude amplification by modeling each Grover step as a CPTP perturbation of the ideal iterate. In

the Liouville representation, the dominant eigenvalues of the noisy superoperator take the perturbative form

$$\lambda_{\pm}(p) = (1 - cp + O(p^2))e^{\pm i(\theta + O(p))}, \tag{5.1}$$

yielding a damped-rotation picture in which coherent amplification occurs on the scale $1/\theta$ while noise suppresses the dynamics on the scale $1/p$.

The competition between these effects determines the optimal stopping time

$$T^* \asymp \min\left(\frac{1}{\theta}, \frac{1}{p}\right), \tag{5.2}$$

and identifies the threshold

$$p \sim \frac{1}{\sqrt{N}}. \tag{5.3}$$

For $p \ll 1/\sqrt{N}$, the Grover mechanism remains spectrally intact up to damping, preserving $O(\sqrt{N})$ scaling. For $p \gtrsim 1/\sqrt{N}$, contraction limits the usable depth and the dynamics becomes noise dominated. This shows that the breakdown of quadratic speedup is governed by spectral contraction of the Grover eigenmodes.

References

[1] L. K. Grover, "A fast quantum mechanical algorithm for database search," Proceedings of the 28th Annual ACM Symposium on Theory of Computing (STOC), 1996.

[2] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018.

[3] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 2000.

[4] L. N. Trefethen and M. Embree, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*, Princeton University Press, 2005.

[5] A. M. Childs, A. Gilyén, Y. Su, T. Low, and N. Wiebe, "Quantum singular value transformation and beyond," SIAM Journal on Computing 50(6), 2021.

[6] G. H. Low and I. L. Chuang, "Hamiltonian simulation by qubitization," Quantum 3, 163 (2019).